

Math 170S - Homework 6 - due February 27

Problem 1. A researcher claims she can reduce the variance of $N(\mu, 100)$ by a new manufacturing process. If S^2 is the variance of a random sample of size n from this new distribution, she tests $H_0 : \sigma^2 = 100$ against $H_1 : \sigma^2 < 100$ by rejecting H_0 if $(n-1)S^2/100 \leq \chi^2_{1-\alpha}(n-1)$ (since $(n-1)S^2/100$ is distributed like $\chi^2(n-1)$ when H_0 is true).

- What is test statistic the researcher uses to test the variance? What is the decision rule? What is null hypothesis? What is alternative hypothesis?
- If $n = 23$, $s^2 = 32.52$, $\alpha = 0.025$, would she reject H_0 ?
- How would you change the decision rule to test the same H_0 against the alternative hypothesis $H_1 : \sigma^2 \neq 100$.

Problem 2. 8.2-6

Problem 3. 8.2-10

Problem 4. 8.3-2 in the red book/8.4-2 in the blue book (it is the same problem in the section about proportions)

Problem 5. 8.3-10 in the red book/8.4-10 in the blue book (it is the same problem in the section about proportions)

Problem 6. 8.4-3 in the red book/8.5-3 in the blue book (same problem about sign and Wilcoxon tests, “t test” is test for the equality for the means from the section 8.1)

* “the blue book” – 10th edition; “the red book” – 9th edition of Hogg, Tanis, Zimmerman